

Network Fundamentals, Nmap & Vulnerability Analysis

Practical Lab Guide & Methodology for Red Team / SOC

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Environment: Kali Linux, Metasploitable2, DVL, VirtualBox

Tools: Nmap, Netcat, Telnet, Vulscan, Vulners

1. Network Architecture Fundamentals & Protocols

A solid understanding of network fundamentals is an essential foundation for any cybersecurity professional. This section synthesizes reference models, key protocols, and methodologies observed during network security audits.

1.1 Reference Models: OSI vs. TCP/IP

The OSI model (7 layers) provides a standardized conceptual framework, whereas the TCP/IP model (4 layers) represents the practical implementation used in global communications.

OSI Layer	Primary Function	TCP/IP Equivalent	Protocols / Examples
7. Application	User interface and network services.	Application	HTTP, HTTPS, SSH, FTP, DNS, SMTP
6. Presentation	Translation, formatting, compression, and encryption.	Application	SSL/TLS, JPEG, MP4
5. Session	Dialogue control and management between applications.	Application	RPC, NetBIOS, Sockets
4. Transport	Flow control, segmentation, and reassembly.	Transport	TCP (Reliable), UDP (Fast)
3. Network	Logical addressing and routing (Hops).	Internet	IP (IPv4/IPv6), ICMP, IPsec, Routers
2. Data Link	Physical addressing (MAC) and framing.	Network Access	Ethernet, Wi-Fi (802.11), ARP, Switches
1. Physical	Binary transmission over physical/wireless media.	Network Access	UTP Cables, Fiber, Hubs, Antennas, Bits

1.2 Transport Protocols: TCP vs. UDP

- **TCP (Transmission Control Protocol):** Connection-oriented via a 3-Way Handshake (SYN → SYN-ACK → ACK). Provides guaranteed delivery, lost packet retransmission, and flow control. Used by HTTP, SSH, SMB, FTP.
- **UDP (User Datagram Protocol):** Connectionless. Sends datagrams without prior verification or acknowledgment. Minimizes latency for real-time transmissions such as VoIP, video streaming, and DNS queries.

1.3 ARP Protocol and MAC/IP Resolution

The Address Resolution Protocol (ARP) acts as the essential bridge between Layer 3 (IP) and Layer 2 (MAC). It enables a host to discover the physical address (MAC) corresponding to an IP within the same local network segment via broadcast requests (ARP Request) and unicast replies (ARP Reply).

```
# Inspect local ARP cache table in Linux/Windows
arp -a
```

```
Home
(root@RedRaptor-Hunt)-[~]
# arp -a
? (10.0.2.2) at 08:00:27:ed:e5:96 [ether] on eth0
? (10.0.2.1) at 52:55:0a:00:02:01 [ether] on eth0
```

2. Network Analysis Methodology with Nmap

Nmap (Network Mapper) is the foundational tool in Active Reconnaissance and Enumeration phases. It maps the attack surface, identifies open ports, discovers software versions, and detects operating systems.

2.1 TCP/IP Flags and Scan Types

Scan Type	Nmap Command	Mechanism & Tactical Advantages
TCP SYN Scan (Stealth)	nmap -sS <target>	Sends a SYN packet. If SYN/ACK is received, sends RST to close the connection without completing it. Fast, stealthy, and avoids flooding application logs. Requires root privileges.
TCP Connect Scan	nmap -sT <target>	Completes the full TCP 3-Way Handshake. Used when the execution environment lacks root/sudo privileges.
Ping Sweep (Discovery)	nmap -sn <network>	Disables port scanning. Performs active host discovery using ICMP requests, ARP (on local LAN), and TCP ACK probes.
UDP Scan	nmap -sU <target>	Audits UDP ports. Slow due to ICMP port-unreachable responses, but critical; services like DNS (53), SNMP (161), and DHCP (67) run on UDP.
ACK Scan (Firewall Rules)	nmap -sA <target>	Evaluates firewall rule filtering. An unfiltered response indicates a Stateless firewall; filtered indicates a Stateful firewall.

2.2 Professional Pentesting Workflow

1. Stealthy Host Discovery (Ping Sweep):

```
nmap -sn -n 10.0.2.0/24
```

```
(root@RedRaptor-Hunt)-[~]
# nmap -sn -n 10.0.2.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-15 22:14 -0700
Nmap scan report for 10.0.2.1
Host is up (0.00085s latency).
MAC Address: 52:55:0A:00:02:01 (Unknown)
Nmap scan report for 10.0.2.2
Host is up (0.00095s latency).
MAC Address: 08:00:27:ED:E5:96 (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.4
Host is up (0.0012s latency).
MAC Address: 08:00:27:04:B1:AD (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.17
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.22 seconds
```

Identifies active IP addresses without triggering alerts caused by massive port scans.

2. Fast Open Port Scan:

```
sudo nmap -sS -Pn -n --top-ports 100 -T4 10.0.2.4
```

```
(root@RedRaptor-Hunt)-[~]
# sudo nmap -sS -Pn -n --top-ports 100 -T4 10.0.2.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-15 22:20 -0700
Nmap scan report for 10.0.2.4
Host is up (0.0054s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 08:00:27:04:B1:AD (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
```

Uses -Pn to bypass ICMP ping blocks, and -n to prevent DNS reverse lookup logging.

3. Comprehensive Service and OS Enumeration:

```
sudo nmap -sS -sV -O -p- -T4 10.0.2.4 -oN full_scan_Metasploitable2.txt
```

Scans all 65,535 ports (-p-), identifies exact software versions (-sV), and performs OS fingerprinting (-O).

3. Manual Banner Grabbing & Service Verification

A golden rule in Red Teaming is manual verification. Automated scanner outputs should not be trusted blindly; verifying whether an exposed service matches actual protocol behavior or is a trap (Honeypot/Banner Spoofing) is mandatory.

3.1 Banner Extraction using Netcat and Telnet

Services such as SSH (port 22) or Web/VNC servers automatically transmit banner identification strings upon establishing a TCP connection.

```
# Manual SSH verification on port 22 using Netcat
nc -vn 10.0.2.4 22
```

```
# Expected output: SSH-1.99-OpenSSH_4.4
```

```
# Manual verification using Telnet  
telnet 10.0.2.4 22
```

```
(root@RedRaptor-Hunt)-[~/New_project_july_26/script_8.2026/reportes_recon]  
# nc -vn 10.0.2.4 22  
(UNKNOWN) [10.0.2.4] 22 (ssh) open  
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

```
(root@RedRaptor-Hunt)-[~/New_project_july_26/script_8.2026/reportes_recon]  
# telnet 10.0.2.4 22  
Trying 10.0.2.4 ...  
Connected to 10.0.2.4.  
Escape character is '^]'.  
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

4. Vulnerability Search Engines: Vulners & Vulscan

Nmap expands its scope from network scanner to vulnerability assessment scanner using Lua scripts via the Nmap Scripting Engine (NSE) located in /usr/share/nmap/scripts/.

4.1 Vulners.nse Script (Cloud API Based)

Queries the Vulners.com API in real time to match software versions gathered via -sV against known CVE databases and CVSS severity ratings.

```
# Executing Vulners script  
nmap -sV --script vulners.nse 10.0.2.4 -oN vulners_results_08.17.26.txt
```

```

(root@RedRaptor-Hunt)-[~/New_project_july_26]
# nmap -sV --script vulners.nse 10.0.2.4 -oN vulners_results_08.17.26.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 10:27 -0700
Nmap scan report for 10.0.2.4
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| vulners:
| vsftpd 2.3.4:
| PACKETSTORM:162145 10.0 https://vulners.com/packetstorm/PACKETSTORM:162145 *EXPLOIT*
| MSF:EXPLOIT-UNIX-FTP-VSFTPD_234_BACKDOOR- 10.0 https://vulners.com/metasploit/MSF:EXPLOIT-UNIX-FTP-V
SFTPD_234_BACKDOOR- *EXPLOIT*
| FEA24DDE-72FF-5445-B198-9E2D5322DC86 10.0 https://vulners.com/githubexploit/FEA24DDE-72FF-5445-B198-9E2
D5322DC86 *EXPLOIT*
| EDB-ID:49757 10.0 https://vulners.com/exploitdb/EDB-ID:49757 *EXPLOIT*
| EDB-ID:17491 10.0 https://vulners.com/exploitdb/EDB-ID:17491 *EXPLOIT*
| ED5791E3-8935-5632-A403-CB7E7FA26CB0 10.0 https://vulners.com/githubexploit/ED5791E3-8935-5632-A403-CB7
E7FA26CB0 *EXPLOIT*
| EC73033B-7726-59C3-828B-D828B8955DB9 10.0 https://vulners.com/githubexploit/EC73033B-7726-59C3-828B-D82
BBB955DB9 *EXPLOIT*
| E9B0AEBB-5138-50BF-8922-2D87E3C046DD 10.0 https://vulners.com/githubexploit/E9B0AEBB-5138-50BF-8922-2D8
7E3C046DD *EXPLOIT*
| E356CCEC-7C81-5A77-8CE4-87DF4DB21897 10.0 https://vulners.com/githubexploit/E356CCEC-7C81-5A77-8CE4-87D
F4DB21897 *EXPLOIT*
| DC8CF54B-5E0D-55F4-B5C1-920406A0FF9F 10.0 https://vulners.com/githubexploit/DC8CF54B-5E0D-55F4-B5C1-920
406A0FF9F *EXPLOIT*
| D0DBA05F-750F-5F9B-9C69-65E8FA6977D5 10.0 https://vulners.com/githubexploit/D0DBA05F-750F-5F9B-9C69-65E
8FA6977D5 *EXPLOIT*
| CVE-2011-2523 10.0 https://vulners.com/cve/CVE-2011-2523
| CNVD-2020-46837 10.0 https://vulners.com/cnvd/CNVD-2020-46837
| CC3F6C15-182F-53F6-A5CC-812D37F1F047 10.0 https://vulners.com/githubexploit/CC3F6C15-182F-53F6-A5CC-812
D37F1F047 *EXPLOIT*
| BEE67BEA-45C2-50C0-88F3-9A2E7CC2DECE 10.0 https://vulners.com/githubexploit/BEE67BEA-45C2-50C0-88F3-9A2
E7CC2DECE *EXPLOIT*
| A7A90EE8-8AB4-5A6C-B232-E81EB21F6397 10.0 https://vulners.com/githubexploit/A7A90EE8-8AB4-5A6C-B232-E81
EB21F6397 *EXPLOIT*
| A418DBAB-C86F-56F2-B9E8-709503C4D553 10.0 https://vulners.com/githubexploit/A418DBAB-C86F-56F2-B9E8-709
503C4D553 *EXPLOIT*
| A41B5EAD-1A4C-56A6-97C6-1C58A1CF1E3B 10.0 https://vulners.com/githubexploit/A41B5EAD-1A4C-56A6-97C6-1C5
8A1CF1E3B *EXPLOIT*
| 9B4BA341-1CBF-59D6-AA40-33596997F23B 10.0 https://vulners.com/githubexploit/9B4BA341-1CBF-59D6-AA40-335
96997F23B *EXPLOIT*
| 9150DFCF-4BDD-5D2E-B527-5E5AC292A4A9 10.0 https://vulners.com/githubexploit/9150DFCF-4BDD-5D2E-B527-5E5
AC292A4A9 *EXPLOIT*
| 885D9854-A37F-5A73-9F90-EE3F205DE5B0 10.0 https://vulners.com/githubexploit/885D9854-A37F-5A73-9F90-EE3
F205DE5B0 *EXPLOIT*
| 88054789-E1E3-5DA3-86E4-387A3D4ACCA8 10.0 https://vulners.com/githubexploit/88054789-E1E3-5DA3-86E4-387
A3D4ACCA8 *EXPLOIT*
| 817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3 10.0 https://vulners.com/githubexploit/817CD8FE-87C4-5FE8-B9B2-8CC
64333A3F3 *EXPLOIT*
| 640E6781-DC00-5CC2-88A5-C12AFFCA478B 10.0 https://vulners.com/githubexploit/640E6781-DC00-5CC2-88A5-C12
AFFCA478B *EXPLOIT*
| 63A5C9A7-C241-5E83-9EE6-ABAB44BDD270 10.0 https://vulners.com/githubexploit/63A5C9A7-C241-5E83-9EE6-ABA
B44BDD270 *EXPLOIT*
| 5F4BCEDE-77DF-5D54-851A-0AE8B76458D9 10.0 https://vulners.com/githubexploit/5F4BCEDE-77DF-5D54-851A-0AE
8B76458D9 *EXPLOIT*
| 59BAFDCD-5027-5C82-BC59-FC7625DD81DB 10.0 https://vulners.com/githubexploit/59BAFDCD-5027-5C82-BC59-FC7
625DD81DB *EXPLOIT*
| 578A4185-29DB-5E4B-9A4B-3C01D25EB897 10.0 https://vulners.com/githubexploit/578A4185-29DB-5E4B-9A4B-3C0
1D25EB897 *EXPLOIT*

```

```

# To verify the contents in the File
ls

```

```

(root@RedRaptor-Hunt)-[~/New_project_july_26]
# ls
2.resol-vulners-dvl.txt full_scan_Metasploitable2.txt resol-vulners-dvl.txt vulners_results_08.17.26.txt

```

```

# To verify the contents of the created text file
cat vulners_results_08.17.26.txt

```



```
(root@RedRaptor-Hunt)-[~/New_proyect_july_26]
# cat vulners_results_08.17.26.txt
# Nmap 7.99 scan initiated Mon Aug 17 10:27:07 2026 as: /usr/lib/nmap/nmap -sV --script vulners.nse -oN vulners_resul
ts_08.17.26.txt 10.0.2.4
Nmap scan report for 10.0.2.4
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| vulners:
|   vsftpd 2.3.4:
|     PACKETSTORM:162145      10.0      https://vulners.com/packetstorm/PACKETSTORM:162145      *EXPLOIT*
|     MSF:EXPLOIT-UNIX-FTP-VSFTPD_234_BACKDOOR- 10.0      https://vulners.com/metasploit/MSF:EXPLOIT-UNIX-FTP-V
SFTPD_234_BACKDOOR- *EXPLOIT*
|     FFA24DDE-72FE-5445-B198-9F2D5322DC86 10.0      https://vulners.com/githubexploit/FFA24DDE-72FE-5445-B198-9F2
```

4.2 Vulscan.nse Script (Local / Offline Analysis)

Unlike Vulners, vulscan executes offline correlations locally against exported CSV databases (CVE, ExploitDB, NVD, SecurityFocus). This provides an ideal solution for auditing isolated or Air-Gapped networks.

```
# Installation and local integration in Kali Linux
cd /usr/share/nmap/scripts/
sudo git clone https://github.com/scipag/vulscan.git
sudo nmap --script-updatedb

# Running Vulscan against local ExploitDB database
nmap -sV --script=vulscan/vulscan.nse --script-args vulscandb=exploitdb.csv 10.0.2.4
```

```
(root@RedRaptor-Hunt)-[~/New_proyect_july_26]
# nmap -sV --script=vulscan/vulscan.nse --script-args vulscandb=exploitdb.csv 10.0.2.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 11:38 -0700
Nmap scan report for 10.0.2.4
Host is up (0.00079s latency).
Not shown: 977 closed tcp ports (reset)
Bug in vulscan: no string output.
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| vulscan: exploitdb.csv:
| [17491] VSFTPD 2.3.4 - Backdoor Command Execution
| [16270] vsftpd 2.3.2 - Denial of Service Vulnerability
|
|_
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| vulscan: exploitdb.csv:
| [21402] OpenSSH 2.x/3.x Kerberos 4 TGT/AFS Token Buffer Overflow Vulnerability
| [3303] Portable OpenSSH ≤ 3.6.1p-PAM / 4.1-SUSE Timing Attack Exploit
| [2444] OpenSSH ≤ 4.3 p1 (Duplicated Block) Remote Denial of Service Exploit
|
|_
23/tcp    open  telnet       Linux telnetd
| vulscan: exploitdb.csv:
| [30902] Linux Kernel 2.6.22 IPv6 Hop-By-Hop Header Remote Denial of Service Vulnerability
| [30605] Linux Kernel 2.6.x ALSA snd-page-alloc Local Proc File Information Disclosure Vulnerability
| [30604] Linux Kernel 2.6.x Ptrace Local Privilege Escalation Vulnerability
| [30469] Red Hat CloudForms Management Engine 5.1 - agent/linuxpkgs Path Traversal
| [30080] Linux Kernel 2.6.x VFat Compat IOCTLS Local Denial of Service Vulnerability
| [29916] Linux Kernel 2.6.x NETLINK_FIB_LOOKUP Local Denial of Service Vulnerability
| [29826] Linux Kernel 2.6.x AppleTalk ATalk_Sum_SKB Function Denial of Service Vulnerability
| [29781] Linux Kernel 2.6.x - IPv6_SockGlue.c NULL Pointer Dereference Vulnerability
| [29714] Linux Kernel 2.6.17 - Sys_Tee Local Privilege Escalation Vulnerability
| [29683] Linux Kernel 2.6.x Audit Subsystems Local Denial of Service Vulnerability
| [29436] MIPS Little Endian Reverse Shell Shellcode (Linux)
| [28912] Linux Kernel 2.6.x ISO9660 Denial of Service Vulnerability
| [28895] Linux Kernel 2.6.x SquashFS Double Free Denial of Service Vulnerability
| [28679] Evince PDF Reader - 2.32.0.145 (Windows) and 3.4.0 (Linux) - Denial of Service
```

5. Practical Case Study & Laboratory Analysis

Isolated Test Environment (Lab Setup)

The assessments described below were conducted within an isolated virtual environment on VirtualBox using segmented NAT Networks composed of Kali Linux (Attacker), Metasploitable2, and DVL (Damn Vulnerable Linux).

5.1 Traceroute Path Analysis & Elevated Latency Findings

During route tracing toward external targets, hop analysis revealed structural anomalies:

```
traceroute google.es
```

```
(root@RedRaptor-Hunt)-[~]
# traceroute google.es
traceroute to google.es (142.251.35.99), 30 hops max, 60 byte packets
 1 192.168.0.1 (192.168.0.1) 488.129 ms 516.630 ms 533.638 ms
 2 192.168.5.1 (192.168.5.1) 519.487 ms 519.683 ms 519.729 ms
 3 192.168.230.1 (192.168.230.1) 545.007 ms 544.904 ms 548.555 ms
 4 192.168.250.1 (192.168.250.1) 548.482 ms 548.412 ms 548.347 ms
 5 172.31.101.1 (172.31.101.1) 549.902 ms 548.214 ms 548.151 ms
 6 10.200.0.61 (10.200.0.61) 578.919 ms 66.384 ms 18.611 ms
 7 131.108.236.9 (131.108.236.9) 38.580 ms 43.687 ms 100.508 ms
 8 * 131.108.236.227 (131.108.236.227) 100.381 ms 82.562 ms
 9 192.178.249.13 (192.178.249.13) 114.416 ms 216.239.62.245 (216.239.62.245) 91.025 ms *
10 142.250.60.142 (142.250.60.142) 86.362 ms 142.250.224.250 (142.250.224.250) 99.757 ms 216.239.50.108 (216.239.50.108) 81.942 ms
11 lcmiaa-an-in-f3.1e100.net (142.251.35.99) 99.426 ms 172.253.69.129 (172.253.69.129) 85.602 ms 81.192 ms
```

Key Security Findings (SOC / Red Team Analysis)

- **Elevated Local Gateway Latency (~140ms):** Standard LAN round-trip times are typically <5ms. Latencies exceeding 100ms on the first hop confirm the presence of an encapsulated VPN tunnel, intermediate proxy, or bandwidth-throttled virtual adapter.
- **Multi-hop & Subnet Segmentation:** Transitioning across Class C (192.168.x.x), Class B (172.31.x.x), and Class A (10.200.x.x) private networks demonstrates multiple layers of NAT and virtual routing prior to internet egress.
- **Packet Filtering at Hop 9 (* *):** Asterisks indicate ICMP Time Exceeded packet drops at the perimeter router, confirming infrastructure hardening.

5.2 Critical Vector Identification on Vulnerable Targets

Port scanning across the laboratory network segment revealed multiple exposed critical services:

Target IP	Port / Service	Software / Version	Risk Level	Attack Vector / Vulnerability
10.0.2.14	22/TCP - SSH	OpenSSH 4.4 (Protocol 1.99)	CRITICAL	Obsolete ciphers, memory corruption/overflow vulnerabilities, and susceptible to brute-force attacks.
10.0.2.14	80/TCP - HTTP	Apache 1.3.37	CRITICAL	Unmaintained web server, exposed directory indexing, and known Remote Code Execution (RCE) flaws.
192.168.1.128	23/TCP - Telnet	BusyBox Telnetd (IoT)	HIGH	Unencrypted cleartext transmission of credentials. Highly vulnerable to network sniffing/MiTM.
10.0.2.4	445/TCP - SMB	Samba 3.0.20	CRITICAL	Vulnerable to Remote Command Execution (Command Injection via Username / EternalBlue scope).

6. Automation: Master Reconnaissance Bash Script

```
# To verify the contents in the File
ls
```

```
# create a folder for the scripts
mkdir
```

```
# To verify the contents in the File
ls
```

```
# enter a the File
cd
```

```
(root@RedRaptor-Hunt)~]
# ls
Desktop    Downloads  Metalbin   New_proyect_july_26  Pictures  Templates
Documents  Home_2026  Music      old_class            Public    Videos

(root@RedRaptor-Hunt)~]
# cd New_proyect_july_26

(root@RedRaptor-Hunt)~/New_proyect_july_26]
# ls
2.resol-vulners-dvl.txt  full_scan_Metasploitable2.txt  resol-vulners-dvl.txt  vulners_results_08.17.26.txt

(root@RedRaptor-Hunt)~/New_proyect_july_26]
# mkdir script_8.2026

(root@RedRaptor-Hunt)~/New_proyect_july_26]
# ls
2.resol-vulners-dvl.txt      resol-vulners-dvl.txt  vulners_results_08.17.26.txt
full_scan_Metasploitable2.txt  script_8.2026

(root@RedRaptor-Hunt)~/New_proyect_july_26]
# cd script_8.2026
```

```
# create the scripts
vim
```

To streamline repetitive audit procedures during lab evaluations and security assessments, the following custom Bash script named `auto-recon.sh` was developed:

```
#!/bin/bash
# =====
# Name: auto-recon.sh
# Date: 08.17.26
# Description: Automated reconnaissance script using Nmap, Vulners, and Vulscan
# Author: Juan Andrés Alba Peralta
# =====

TARGET=$1

if [ -z "$TARGET" ]; then
```



```

    echo -e "
[!] Usage: $0 <Target-IP-or-Range>
    echo -e "[*] Example: $0 10.0.2.18
"
    exit 1
fi

REPORT_DIR="./reportes_recon"
mkdir -p $REPORT_DIR

echo -e "
[+] Starting Phase 1: Port and Service Discovery on $TARGET..."
sudo nmap -sS -sV -O -Pn -T4 -p- $TARGET -oN $REPORT_DIR/nmap_full_$TARGET.txt

echo -e "
[+] Starting Phase 2: Vulnerability Audit (Vulners API & Local Vulscan)..."
sudo nmap -sV --script vulners.nse,vulscan/vulscan.nse $TARGET -oN
$REPORT_DIR/vulnerabilities_$TARGET.txt

echo -e "
[✓] Reconnaissance completed successfully."
echo -e "[*] Reports saved to directory: $REPORT_DIR
"

```

```

root@RedRaptor-Hunt: ~/New_proyect_july_26/script_8.2026
Session Actions Edit View Help
1 #!/bin/bash
2 # Name: auto-recon.sh
3 # Date: 08.17.26
4 # Description: Automated reconnaissance script using Nmap, Vulners, and Vulscan
5 # Author: Juan Andrés Alba Peralta
6 #
7
8 TARGET=$1
9
10 if [ -z "$TARGET" ]; then
11     echo -e "
12 [!] Usage: $0 <Target-IP-or-Range>
13     echo -e "[*] Example: $0 10.0.2.18
14 "
15     exit 1
16 fi
17
18 REPORT_DIR="./reportes_recon"
19 mkdir -p $REPORT_DIR
20
21 echo -e "
22 [+] Starting Phase 1: Port and Service Discovery on $TARGET ..."
23 sudo nmap -sS -sV -O -Pn -T4 -p- $TARGET -oN $REPORT_DIR/nmap_full_$TARGET.txt
24
25 echo -e "
26 [+] Starting Phase 2: Vulnerability Audit (Vulners API & Local Vulscan)..."
27 sudo nmap -sV --script vulners.nse,vulscan/vulscan.nse $TARGET -oN $REPORT_DIR/vulnerabilities_$TARGET.txt
28
29 echo -e "
30 [✓] Reconnaissance completed successfully."
31 echo -e "[*] Reports saved to directory: $REPORT_DIR
32 "
33

```

```

# To verify the contents in the File
ls

```

```
# make it executable  
chmod
```

```
# To verify the script has permissions  
ls -la
```

```
(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026]  
# ls  
auto-recon.sh  
  
(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026]  
# chmod +x auto-recon.sh  
  
(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026]  
# ls -la  
total 12  
drwxr-xr-x 2 root root 4096 Aug 17 14:03 .  
drwxr-xr-x 3 root root 4096 Aug 17 13:27 ..  
-rwxr-xr-x 1 root root 986 Aug 17 14:03 auto-recon.sh  
  
(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026]  
#
```

```
# To verify the scripts  
./
```

```

(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026]
# ./auto-recon.sh 10.0.2.4

[+] Starting Phase 1: Port and Service Discovery on 10.0.2.4 ...
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 14:14 -0700
Nmap scan report for 10.0.2.4
Host is up (0.0013s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
42909/tcp open  mountd       1-3 (RPC #100005)
56867/tcp open  status       1 (RPC #100024)
57043/tcp open  nlockmgr     1-4 (RPC #100021)
57933/tcp open  java-rmi     GNU Classpath grmiregistry
MAC Address: 08:00:27:04:B1:AD (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 165.18 seconds

[+] Starting Phase 2: Vulnerability Audit (Vulners API & Local Vulscan)...
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 14:16 -0700
| [1001115] ASPSeek CGI-based Search Engine May Execute Arbitrary Code Supplied by Remote Users
| [1001083] Apache Web Server May Display Directory Index Listings Even if Directory Listings Are Disabled
| [1000942] Resin Web Servlet and Java Engine Allows Unauthorized Access to Directories and Files Outside of the Web Root Directory
|
| OSVDB - http://www.osvdb.org:
| [82782] Apache CXF WS-SecurityPolicy 1.1 SupportingToken Policy Bypass
| [90864] Apache Batik 1xx Redirect Script Origin Restriction Bypass
| [73550] Foxit Reader FreeType Engine Type 1 Font Decoder Overflow
| [6630] Apache Tomcat Java Server Pages (JSP) Engine WPrinterJob() DoS
|
MAC Address: 08:00:27:04:B1:AD (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 431.10 seconds

[v] Reconnaissance completed successfully.
[*] Reports saved to directory: ./reportes_recon

```

```

# To verify the script report
ls

```

```
cd
```

```
(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026]  
# ls  
auto-recon.sh  reportes_recon
```

```
(root@RedRaptor-Hunt)-[~/New_proyect_july_26/script_8.2026/reportes_recon]  
# ls  
nmap_full_10.0.2.4.txt  vulnerabilities_10.0.2.4.txt
```